Integrating Polygenic Scores with Clinical, Lifestyle, and Social Risk Factors to Improve Heart Failure Risk Prediction

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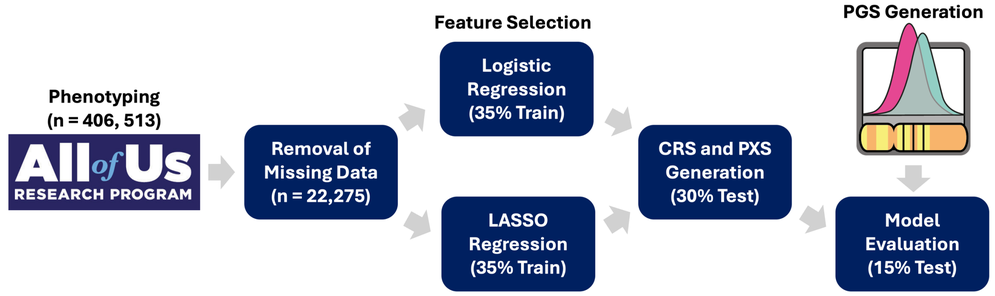
Introduction

- Heart failure (HF) is highly prevalent, high-burden disorder with rising prevalence[1].
- Early HF detection improves outcomes, highlighting the need for new methods[1].
- There are many well-known, non-genomic risk factors of HF, in the clinical, lifestyle, and social determinant of health (SDOH) domains, and it is unclear how genetic and non-genetic risk factors collectively contribute to HF risk[2].
- HF has a heritability of 34% and Polygenic scores (PGS) predict HF[3,4].
- To address this question, we assessed whether combining HF PGS with clinical, lifestyle, and SDOH risk factors improves risk prediction.

Methods



$$PGS = \sum_{j=1}^{m} X_j \beta_j$$

$$CRS = \sum_{i=1}^{n} X_i \beta_i$$

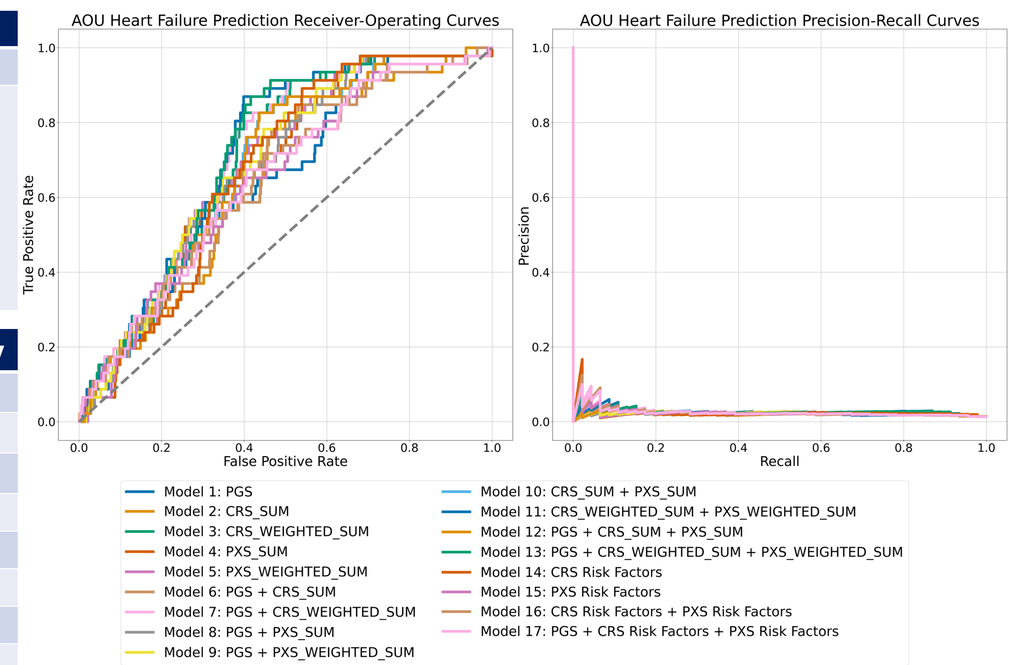
$$PXS = \sum_{k=1}^{o} X_k \beta_k$$

$$IRM: logit(HF) \sim PGS + CRS + PXS + AGE + SEX$$

Results

Clinical Risk Score (CRS)	Polyexposure Score (PXS)	
Clinical	**Lifestyle**	**Social Determinants of Health**
1. Diastolic* and Systolic Blood Pressure*	1. BMI	1. Census-based income
2. Glucose*	2. Nutrition status*	2. Education level
3. HbA1c*	3. Physical activity	3. Income level
4. HDL and LDL cholesterol	4. Smoking status	4. Neighborhood*
5. T2D status		5. Single-level status*
6. Triglycerides*		6. Deprivation index*

Risk Factor	AUROC	AUPRC	F1 Score	Balanced Accuracy
PGS	0.669	0.034	0.044	0.607
CRS_SUM	0.676	0.033	0.042	0.598
CRS_WEIGHTED_SUM	0.736	0.042	0.055	0.655
PXS_SUM	0.709	0.032	0.052	0.646
PXS_WEIGHTED_SUM	0.730	0.034	0.056	0.660
PGS + CRS_SUM	0.678	0.036	0.043	0.603
PGS + CRS_WEIGHTED_SUM	0.737	0.046	0.057	0.662
PGS + PXS_SUM	0.715	0.036	0.054	0.652
PGS + PXS_WEIGHTED_SUM	0.734	0.038	0.057	0.665
CRS_SUM + PXS_SUM	0.732	0.039	0.055	0.652
CRS_WEIGHTED_SUM + PXS_WEIGHTED_SUM	0.762	0.043	0.061	0.678
PGS + CRS_SUM + PXS_SUM	0.734	0.041	0.057	0.658
PGS + CRS_WEIGHTED_SUM + PXS_WEIGHTED_SUM	**0.763**	**0.047**	**0.062**	**0.682**
CRS Risk Factors	0.729	0.040	0.055	0.655
PXS Risk Factors	0.723	0.041	0.060	0.647
CRS Risk Factors + PXS Risk Factors	0.738	0.046	0.065	0.658
PGS + CRS risk factors + PXS risk factors	0.738	0.047	0.065	0.657



References and Acknowledgements

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